Week 3 Discussion

Covers: conditional probability, Bayes' rule, independence and random variables. Problems 1 and 3 are adapted from C&B §1.7.

Problem 1 (Noisy telegraph: Bayes' rule and independence; C&B Exercise 1.41)

As in Example 1.3.6, telegraph signals "dot" and "dash" are sent in the proportion $3:4$. Erratic transmissions cause a dot to become a dash with probability $\frac{1}{4}$ and a dash to become a dot with probability $\frac{1}{3}$.

Useful tools: the law of total probability, $P(B) = \sum_i P(B \mid A_i) P(A_i)$, and Bayes' rule (Theorem 1.3.5), $P(A_i \mid B) = P(B \mid A_i) P(A_i) / P(B)$.

(a) Write down $P(\text{dot sent})$, $P(\text{dash sent})$, and the four conditional probabilities $P(\text{received} \mid \text{sent})$.

(b) Find $P(\text{dash received})$. Which partition of S are you using?

(c) If a dash is received, what is the probability that a dash was sent?

(d) Now suppose a two-symbol message is sent. Assume the two symbols are chosen independently, each with the $3:4$ proportions, and that transmission errors happen independently for the two symbols. If the message dot-dot is received, what is the probability distribution of the four possible messages (dot-dot, dot-dash, dash-dot, dash-dash) that could have been sent?

Hint for (d): first find $P(\text{dot sent} \mid \text{dot received})$ and $P(\text{dash sent} \mid \text{dot received})$ for a single symbol. Then explain why independence lets you multiply.

> **(a)** $P(\text{dot sent}) = \frac{3}{7}$ and $P(\text{dash sent}) = \frac{4}{7}$.
>
> | Sent | Received dot | Received dash |
> |---|---|---|
> | dot | $3/4$ | $1/4$ |
> | dash | $1/3$ | $2/3$ |
>
> **(b)** Partition by what was sent, $\{\text{dot sent}, \text{dash sent}\}$:
>
> $$P(\text{dash rec.}) = \tfrac{1}{4}\cdot\tfrac{3}{7} + \tfrac{2}{3}\cdot\tfrac{4}{7} = \tfrac{9}{84} + \tfrac{32}{84} = \tfrac{41}{84}.$$
>
> **(c)**
>
> $$P(\text{dash sent} \mid \text{dash rec.}) = \frac{(2/3)(4/7)}{41/84} = \frac{32/84}{41/84} = \frac{32}{41} \approx 0.780.$$
>
> **(d)** For one symbol, $P(\text{dot rec.}) = 1 - \frac{41}{84} = \frac{43}{84}$, so
>
> $$P(\text{dot sent} \mid \text{dot rec.}) = \frac{(3/4)(3/7)}{43/84} = \frac{27}{43}, \qquad P(\text{dash sent} \mid \text{dot rec.}) = \frac{16}{43}.$$
>
> The pairs (symbol sent, symbol received) for the two positions are independent, so the posteriors multiply:

Sent	Probability given dot-dot received
dot-dot	$(27/43)^2 = 729/1849 \approx 0.394$
dot-dash	$(27 \cdot 16)/43^2 = 432/1849 \approx 0.234$
dash-dot	$432/1849 \approx 0.234$
dash-dash	$(16/43)^2 = 256/1849 \approx 0.138$

The four probabilities sum to $1849/1849 = 1$. Even though dot-dot was received, there is about a 61% chance that at least one symbol was actually a dash.

Problem 2 (Pairwise vs. mutual independence)

Recall Definition 1.3.12: $A_1, \dots, A_n$ are **mutually independent** if the product rule holds for *every* subcollection, not just for pairs.

(a) Roll two fair dice, with all 36 ordered outcomes equally likely. Let A = "the first die is odd," B = "the second die is odd," and C = "the sum is odd." Find $P(A)$, $P(B)$, $P(C)$, and check whether each of the pairs (A, B), (A, C), (B, C) is independent.

(b) Are A, B, C mutually independent? Compute $P(A \cap B \cap C)$ and $P(C \mid A \cap B)$, and explain in words why learning A alone tells you nothing about C, while learning A and B together tells you everything.

(c) How many equations must be checked to show that four events A_1, A_2, A_3, A_4 are mutually independent? List them by size of the subcollection.

(d) A system has three components. Components 1 and 2 are in **parallel** (the pair works if at least one of them works), and that pair is in **series** with component 3 (the system works only if the pair works *and* component 3 works). Let A_i = "component i works," with A_1, A_2, A_3 mutually independent and $P(A_i) = p$.

1. Write the event "the system works" in terms of A_1, A_2, A_3.
2. Show that $P\big(A_3 \cap (A_1 \cup A_2)\big) = P(A_3)\,P(A_1 \cup A_2)$. *Hint:* distribute, then use Theorem 1.2.9(b). Which of your equations needs the **triple** product rule, so that pairwise independence would not be enough?
3. Find $P(\text{system works})$ as a function of p, and evaluate it at $p = 0.9$.

(a) $P(A) = P(B) = \frac{18}{36} = \frac{1}{2}$. The sum is odd exactly when one die is odd and the other is even, so $P(C) = \frac{9+9}{36} = \frac{1}{2}$.

- $A \cap B$: both odd, $\frac{9}{36} = \frac{1}{4} = P(A)P(B)$.
- $A \cap C$: first odd and second even, $\frac{9}{36} = \frac{1}{4} = P(A)P(C)$.
- $B \cap C$: first even and second odd, $\frac{9}{36} = \frac{1}{4} = P(B)P(C)$.

So the three events are **pairwise independent**.

(b) If both dice are odd, the sum is even, so $A \cap B \cap C = \emptyset$ and

$$P(A \cap B \cap C) = 0 \neq \tfrac{1}{8} = P(A)P(B)P(C).$$

They are **not** mutually independent. Also $P(C \mid A \cap B) = 0 \neq P(C)$. Knowing only the first die is odd leaves the sum odd or even with probability $\frac{1}{2}$ each, because it depends on the unknown second die. Knowing the parity of both dice determines the parity of the sum. (Compare Example 1.3.11.)

(c) One equation for every subcollection of size at least 2: $\binom{4}{2} = 6$ pairs, $\binom{4}{3} = 4$ triples, and $\binom{4}{4} = 1$ quadruple, so $6 + 4 + 1 = 11 = 2^4 - 4 - 1$ equations.

(d)

1. $\{\text{system works}\} = A_3 \cap (A_1 \cup A_2)$.
2. By the distributive law, $A_3 \cap (A_1 \cup A_2) = (A_3 \cap A_1) \cup (A_3 \cap A_2)$, and the intersection of these two pieces is $A_1 \cap A_2 \cap A_3$. By Theorem 1.2.9(b),

   $$P\big(A_3 \cap (A_1 \cup A_2)\big) = P(A_1 \cap A_3) + P(A_2 \cap A_3) - P(A_1 \cap A_2 \cap A_3) = p^2 + p^2 - p^3 = p\,(2p - p^2).$$

   Since $P(A_1 \cup A_2) = p + p - p^2 = 2p - p^2$, this equals $P(A_3)P(A_1 \cup A_2)$. The term $P(A_1 \cap A_2 \cap A_3) = p^3$ uses the triple product rule. With only pairwise independence it could be anything (in part (b) it was 0).
3. $P(\text{system works}) = p\,(2p - p^2) = p\,\big(1 - (1-p)^2\big)$. At $p = 0.9$: $0.9 \times 0.99 = 0.891$.

Takeaway: "independent components" in reliability, or "independent tosses" in Example 1.3.13, always means **mutual** independence. That is what lets us treat combinations of some events, like $A_1 \cup A_2$, as independent of the others.

Problem 3 (Guessing on a standardized test: a random variable; C&B Exercise 1.44)

A test has 20 multiple-choice questions, each with 4 possible answers. A student guesses on every question, so the taking of the exam can be modeled as 20 independent events, each "correct" with probability $\frac{1}{4}$.

(a) Describe the sample space S. How many outcomes does it have? Are they equally likely? *Hint:* record each answer as correct (C) or wrong (W).

(b) Let X = the number of questions answered correctly. Explain why X is a random variable in the sense of Definition 1.4.1. What is its range $\mathcal{X}$?

(c) Fix one outcome s with exactly k correct answers, for example $s = \text{CC}\cdots\text{C}\,\text{WW}\cdots\text{W}$. Use independence to find $P(\{s\})$.

(d) Use the induced probability function (1.4.1), $P(X = k) = P(\{s \in S : X(s) = k\})$, together with counting, to show that

$$P(X = k) = \binom{20}{k}\left(\frac{1}{4}\right)^{k}\left(\frac{3}{4}\right)^{20-k}, \qquad k = 0, 1, \ldots, 20.$$

(e) Find the probability that the student gets at least 10 questions correct. Also find $P(X \geq 1)$ using the complement.

(a) S = all strings of length 20 over $\{\mathrm{C}, \mathrm{W}\}$, so $N(S) = 2^{20}$. The outcomes are **not** equally likely, because C has probability $\frac{1}{4}$ and W has probability $\frac{3}{4}$. (We could use the 4^{20} equally likely answer sheets instead, but then we would have to count the ones with k correct answers as $\binom{20}{k}3^{20-k}$, which gives the same result.)

(b) X assigns to each outcome $s \in S$ the real number of Cs in s, so X is a function from S into the real numbers. $\mathcal{X} = \{0, 1, \ldots, 20\}$.

(c) By mutual independence (Definition 1.3.12) and Theorem 1.3.9 for the complements, $P(\{s\}) = \left(\frac{1}{4}\right)^{k}\left(\frac{3}{4}\right)^{20-k}$. Every outcome with exactly k Cs has this same probability, no matter where the Cs are.

(d) $\{X = k\}$ consists of the $\binom{20}{k}$ strings with C in exactly k positions (choose the positions, unordered, without replacement). Each has the probability in (c), and the outcomes are disjoint, so their probabilities add. This is the **binomial distribution**, which we will meet again in Chapter 3.

(e)

$$P(X \geq 10) = \sum_{k=10}^{20} \binom{20}{k}\left(\frac{1}{4}\right)^{k}\left(\frac{3}{4}\right)^{20-k} \approx 0.0139.$$

Guessing passes a "50%" bar only about 1.4% of the time. On the other hand, $P(X \geq 1) = 1 - P(X = 0) = 1 - (3/4)^{20} \approx 0.997$.